Tavano Matteo SM380057

Physics-Informed Neural Network for Lithium-Ion Battery degradation stable modeling and prognosis

Final exam paper of the **Simulation Intelligence** course at **UniTs** - June 2026

Authors: Fujin Wang, Zhi Zhai, Zhibin Zhao, Yi Di, Xuefeng Chen
Xi'an Jiaotong University · School of Mechanical Engineering

SOH Estimation

PINN Architecture

Battery Management

Problem Formulation: Why estimating SOH is non-trivial?

State of Health: Definition

$$SOH = Q_k \;/\; Q_0$$

Q_k = capacity at cycle k Q_0 = nominal capacity

End-of-first life threshold: SOH ≤ 0.80

> Estimating SOH requires integrating discharge curve, not always possible in service batteries for critical infrastructures.

Why direct measurement fails in the field

- Battery is **rarely fully discharged** ⟶ cannot integrate the discharge curve to get Q_k
- Drift accumulates over time ⟶ **measurement error** compounds
- C-rate and temperature alter the apparent extractable capacity ⟶ **biased estimates** without controlled conditions

Classical Models — Structural Limitation

Linear	$u(t) = a - b \cdot t$
Exponential	$u(t) = a \cdot \exp(-b \cdot t)$
Power-law	$u(t) = a \cdot t\text{^}b$
FFM	$u(t) = \text{empirical } f(t)$

All are univariate functions of time only u(t) — ignoring DoD, temperature, C-rate, SoC, SEI growth…

This paper considers u = f(t, x) where x captures multivariate feature vector

Dataset Overview

387 batteries · 310,705 samples · 4 chemistries · run-to-failure

387	310K	3	1
total batteries	training samples	public datasets	custom dataset (**XJTU**)

XJTU

Batteries
55

Chemistry
NCM

Formula
$LiNi_{0.5}Co_{0.2}Mn_{0.3}O_2$

Nominal Capacity
2000 mAh

Cut-off
2.5–4.2 V

Temperature
Room temp.

TJU

Batteries
130

Chemistry
NCA/NCM/Blend

Formula
$LiNiCoAlO_2$ / $LiNiCoMnO_2$

Nominal Capacity
2500–3500 mAh

Cut-off
2.5–4.2 V

Temperature
25 / 35 / 45 °C

MIT

Batteries
125

Chemistry
LFP

Formula
$LiFePO_4$

Nominal Capacity
1100 mAh

Cut-off
2.0–3.6 V

Temperature
30 °C

HUST

Batteries
77

Chemistry
LFP

Formula
$LiFePO_4$

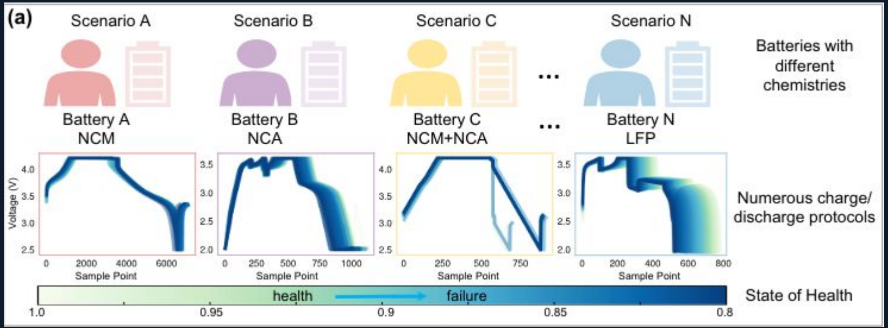
Nominal Capacity
1100 mAh

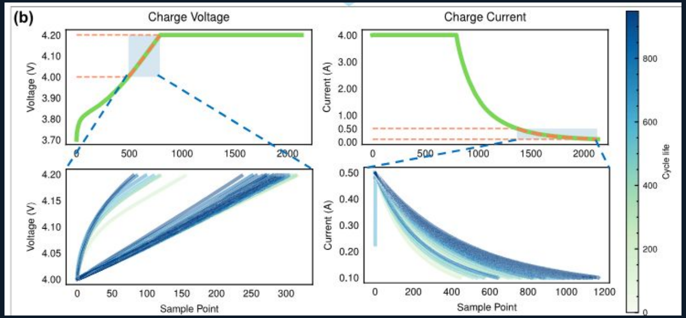
Cut-off
2.0–3.6 V

Temperature
30 °C

Each sample i = one full charge cycle $\longrightarrow (t^i, x^i, u^i)$ · 1 sample per cycle per cell · avg 803 cycles/cell

Feature Extraction from CC-CV Charging Curve





1 Why charge, not discharge?

Discharge is **user-specific**, rarely complete in the field.
Charge is fixed & regular; battery almost always reaches full charge.
CC-CV mode is **universal** across all 4 datasets regardless of protocol.

2 Selection window

Voltage: **V ∈ [Vend − 0.2, Vend] V** *(CC phase)*
Current: **I ∈ [0.1, 0.5] A** *(CV phase)*
This window always exists if the battery is fully charged.

3 16 statistical features ($x \in \mathbb{R}^{16}$)

8 from voltage curve (CC) + 8 from current curve (CV):
Mean · Std · Kurtosis · Skewness ·Charging time · Accumulated charge · Slope · Entropy
Min-max normalized to [−1, 1] ⟶ input to F(·)

Feature Extraction Process: CC-CV curve ⟶ window selection ⟶ 16 statistics ⟶ normalize [−1,1] ⟶ concat with cycle t ⟶ input (t, x) ∈ $\mathbb{R}^{17}$ ⟶ F(·)

Battery Degradation Modeling

SOH — Multivariate function

$$u = f\,(t, x)$$

t = cycle number
x = vector of 16 health indicators from CC-CV
SOH as multivariate function

PDE — SOH decay rate

$$\partial u \,/\, \partial t \;=\; g(t, x, u\,;\,\theta)$$

g(·) is the **nonlinear degradation dynamics** function
Encodes how fast SOH decays given current state & conditions

Approximator — Generalised approximator

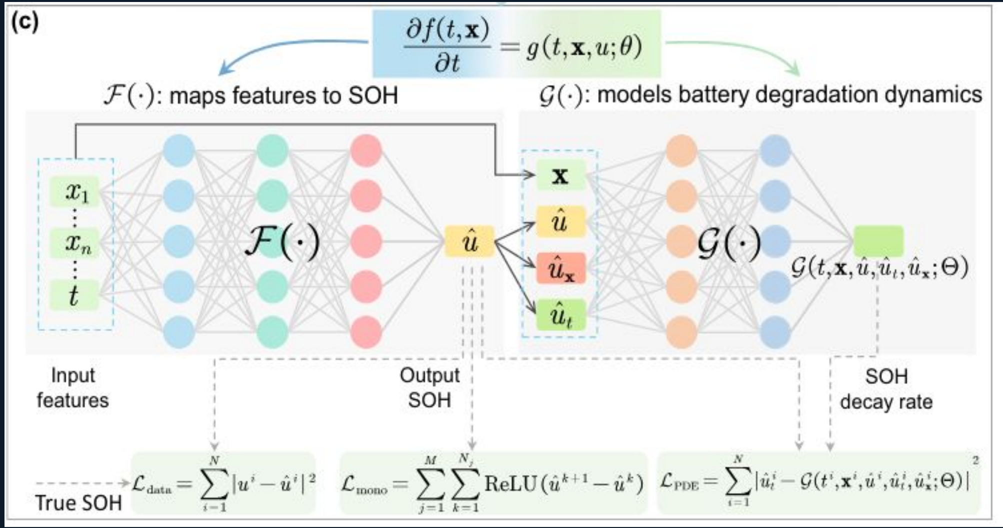
$$u_t \;\approx\; g'(t, x, u, u_t, u_x\,;\,\theta')$$

g(·) is **UNKNOWN** and cannot be derived from first principles
g′(·) is a universal function approximator (neural network G)
Includes first-order partial derivatives $u_t = \partial u/\partial t$, $u_x = \partial u/\partial x$
Higher-order terms discarded for computational tractability

Core challenge: the explicit form of g(·) is driven by complex electrochemical, chemical and physical degradation mechanisms. Therefore, it is learned from data.

PINN Architecture

F(·) — solution network · G(·) — dynamics network · H — physics residual



F(t, x ; Φ) — Solution Network

Maps (t, x) ⟶ û (**estimated SOH**)
Learnable parameters **Φ**
AD computes u□ = ∂û/∂t and ux = ∂û/∂x from this network
Outputs feed into G as additional inputs

G(t, x, u, u□, ux ; Θ) — Dynamics Network

Models the unknown g′(·) — **degradation rate**
Learnable parameters **Θ**
Receives u□ and ux computed via AD from F
Outputs estimated ∂u/∂t (how fast SOH decays)

Total Loss

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \alpha \cdot \mathcal{L}_{\text{PDE}} + \beta \cdot \mathcal{L}_{\text{mono}}$$

$\mathcal{L}_{\text{data}} = \Sigma|u^i - \hat{u}^i|^2$
estimated vs true SOH

$\mathcal{L}_{\text{PDE}} = \Sigma|H(t^i,x^i)|^2$
enforce physics

$\mathcal{L}_{\text{mono}} = \Sigma$ ReLU(û^{k+1}−û^k)
SOH monotone ↓

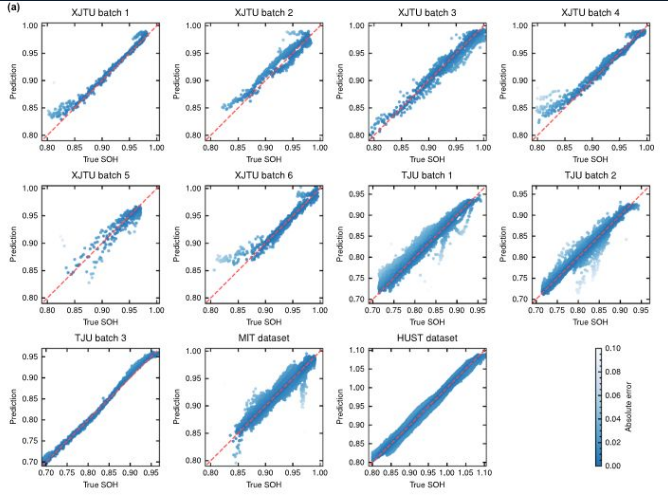
H := ∂F(t,x;Φ)/∂t − G(t, x, u, u□, ux ; Θ) = 0

Physics residual — minimised during training via $\mathcal{L}_{\text{PDE}} = \Sigma\,|H(t^i, x^i)|^2$

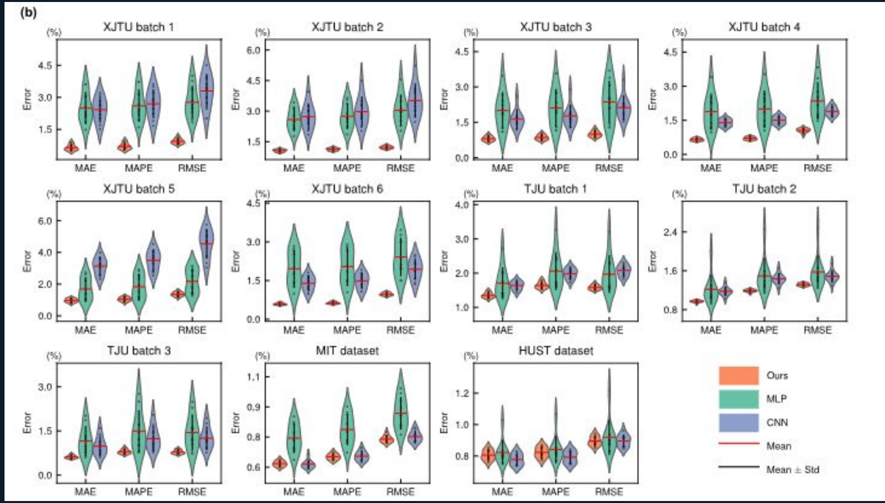
Classical PINN Theory vs This Paper

	Classical PINN (Raissi et al. 2019)	This Paper (Wang et al. 2024)
Governing equation	Known explicitly e.g. $\partial u/\partial t - \alpha\partial^2 u/\partial x^2 = 0$	Unknown, $g(\cdot)$ has no closed form Neural network G approximates $g'(\cdot)$
Number of networks	1 network: $u_\theta(t, x)$ approximates solution	2 networks: $F(\cdot)$ = solution, $G(\cdot)$ = dynamics operator
Problem type	**Forward problem**: solve known PDE or **Inverse problem**: identify known-form parameters	**Operator discovery** and **Parameter learning:** learn the PDE and feature mapping
Physics enforcement	N[u] computed analytically from known PDE $\mathcal{L} = \mathcal{L}_{data} + \cdot\mathcal{L}_{PDE}$	Soft constraint only (H minimised, not imposed) $\mathcal{L} = \mathcal{L}_{data} + \alpha\cdot\mathcal{L}_{PDE} + \beta\cdot\mathcal{L}_{mono}$ +mono = domain-specific physics prior
Input variables x	**Physical spatial coordinates** clear geometric meaning	**16 abstract statistical features** partial derivatives $\partial\hat{u}/\partial x_i$ less interpretable
Transfer learning	Not typically addressed PDE fixed for a given physical system	Explicit mechanism for cross-domain transfer

Results of SOH Estimation



SOH estimation results among all batches Predicted vs actual SOH



MAPE, MAE, RMSE of the three proposed architectures, among all batches, results of 10 runs.

Results of SOH Estimation

Dataset	Batch	Ours		MLP		CNN	
		MAPE	RMSE	MAPE	RMSE	MAPE	RMSE
XJTU	1	**0.0070**	**0.0094**	0.0260	0.0277	0.0270	0.0330
	2	**0.0113**	**0.0122**	0.0275	0.0304	0.0298	0.0352
	3	**0.0086**	**0.0100**	0.0211	0.0237	0.0177	0.0212
	4	**0.0071**	**0.0105**	0.0200	0.0235	0.0150	0.0189
	5	**0.0105**	**0.0135**	0.0183	0.0217	0.0350	0.0453
	6	**0.0063**	**0.0097**	0.0204	0.0242	0.0149	0.0194
TJU	1	**0.0164**	**0.0158**	0.0206	0.0197	0.0198	0.0208
	2	**0.0119**	**0.0132**	0.0149	0.0157	0.0143	0.0149
	3	**0.0080**	**0.0079**	0.0150	0.0144	0.0124	0.0125
MIT		**0.0065**	**0.0074**	0.0079	0.0087	**0.0065**	0.0075
HUST		0.0078	**0.0087**	0.0080	0.0090	**0.0074**	0.0087

PINN avg. MAPE

0.87%

MLP avg. MAPE

1.8%

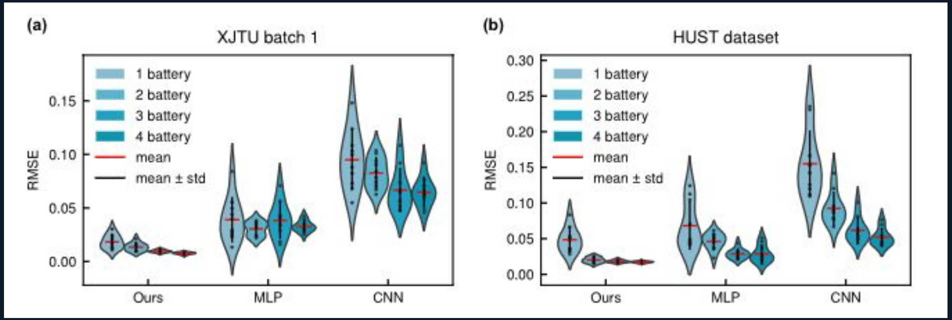
CNN avg. MAPE

2.10%

On data-rich datasets (MIT, HUST): PINN ≈ MLP ≈ CNN
Physics prior **helps most when data is scarce**, not when it is abundant

Stability: PINN shows smallest error variance across 10 independent runs on all datasets, especially on small-batch XJTU where MLP/CNN variate significantly

Results of Small Sample Regime



Dataset	Train batteries	Ours		MLP		CNN	
		MAPE	RMSE	MAPE	RMSE	MAPE	RMSE
XJTU	1	**0.0141**	**0.0184**	0.0343	0.0390	0.0929	0.0949
	2	**0.0105**	**0.0134**	0.0267	0.0304	0.0728	0.0826
	3	**0.0069**	**0.0096**	0.0347	0.0383	0.0548	0.0666
	4	**0.0056**	**0.0076**	0.0292	0.0327	0.0560	0.0647
HUST	1	**0.0446**	**0.0485**	0.0601	0.0682	0.3614	0.1550
	2	**0.0178**	**0.0202**	0.0391	0.0461	0.0826	0.0925
	3	**0.0154**	**0.0181**	0.0251	0.0287	0.0514	0.0618
	4	**0.0144**	**0.0173**	0.0253	0.0288	0.0429	0.0521

PINN with 1 battery ≈ MLP / CNN with 3–4 batteries · Physics prior compensates for data scarcity

Transfer Mechanism

Transfer Mechanism

1. **Pre-train on source domain**

   Train both F(·) and G(·) on full source dataset

2. **Freeze G(·)**

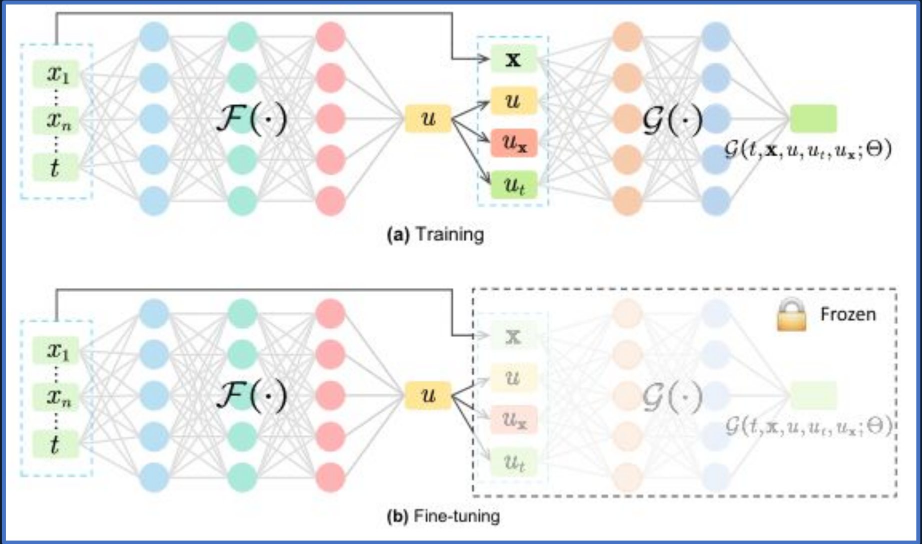
   G(·) parameters Θ fixed, assumed to encode universal degradation dynamics

3. **Fine-tune F(·) on target**

   Update only Φ using 1 battery data from target domain

4. **Test on remaining target cells**

   Evaluate on the fixed test set of target domain



Results of Fine-Tuning Between Datasets

	Fine-tuning				Source-only			
	XJTU	TJU	MIT	HUST	XJTU	TJU	MIT	HUST
XJTU	–	**0.0100**	0.0145	*0.0104*	–	0.0967	0.1329	0.0733
TJU	**0.0093**	–	*0.0119*	0.0146	0.1266	–	0.1674	0.1266
MIT	*0.0239*	0.0272	–	0.0248	0.0347	0.1277	–	0.0561
HUST	0.0333	0.0343	*0.0307*	–	0.1131	0.2008	0.0801	–

Experimental conditions:

- **Fine-tuning**: source pre-train + 1 target battery ⟶ F fine-tuned
- **Source-only**: source pre-train, no target data, direct test

Key Results

- Fine-tuning consistently **outperforms** source-only transfer across all dataset pairs,
- Chemically similar battery pairs (XJTU↔TJU, both NMC) show the **strongest transfer**, while cross-chemistry pairs (XJTU⟶MIT, LFP) Obtains smaller gains
- G(·) captures **chemistry-specific degradation patterns** rather than a fully universal dynamics.

Discussion & Conclusions

Demonstrated

✓ Universal **CC-CV features** work across NCM, NCA, LFP.

✓ **Two-network PINN (F+G)** performs slightly better than MLP/CNN in all experimental settings

✓ **Small-sample**: 1-battery PINN ≈ 3–4 battery MLP. Physics prior acts as a data-efficient regulariser

✓ **Transfer mechanism**: freeze G, fine-tune F with 1 target battery → RMSE is 10 times better than source-only

✓ **XJTU dataset** (55 cells, 6 protocols, run-to-failure) publicly released

Critical Analysis

Did G(·) learn true degradation dynamics?
Cannot confirm mechanistically. 'Universality' is inferred from RMSE, not from inspecting G's outputs against known physics (P2D, SEI models). It may simply act as a well-regularised initialiser.

Is the physics constraint truly enforced?
No. H = 0 is a soft constraint minimised in the loss. The PDE is encouraged, not imposed.

Is the small-sample result robust?
Partially. Tested on only 2/4 datasets in main text. The 10-run average controls variance but cannot rule out dataset-specific advantages.

Is SOH estimation coherent?
Yes, strongly. Scatter plots align tightly with the diagonal across all 11 batches/datasets. MAPE 0.87% is consistent and stable. This is the most credible result in the paper.

Open limits:

! α, β hyperparameters not discussed in main text

! **Chemistry conjecture**: only NiCo-x and LFP tested — untested on LMO, LTO, solid-state

! G(·) **interpretability**: no comparison between frozen and randomly initialized G during transfer learning

Thanks for listening !